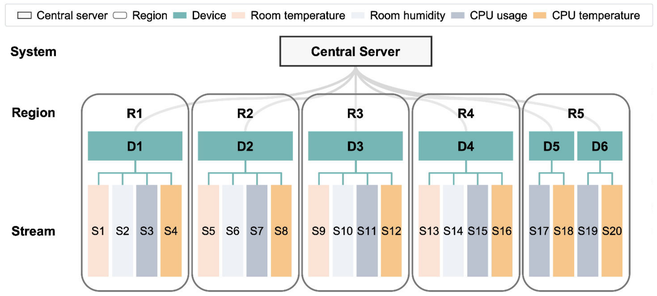
Multi-Agent Reinforcement Learning for Anomaly Detection with Hierarchical Adaptive Sampling

Xin Xia[a], Jie Guo[b], Ziqian Zheng[a], Yujie Ma[b], Chen Zhang[b], Kaibo Liu[a]

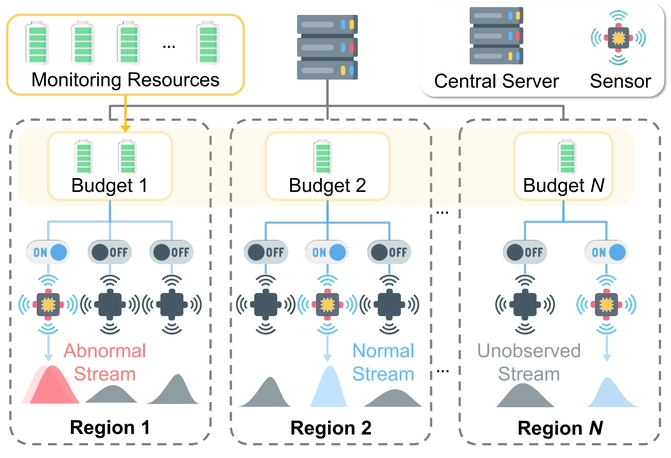
[a]Department of Industrial and Systems Engineering, University of Wisconsin-Madison; [b]Department of Industrial Engineering, Tsinghua University

Motivation

In many practical systems, sensors are grouped into distinct regions.



- **Structure:** System → Regions → Streams (SRS)
- **Real-time Anomaly Detection for SRS systems:**
  1) Hierarchical & decentralized layout
  2) Limited budget for monitoring
  3) Usage → cost
- **Objective:**
  1) Dynamic budget allocation & autonomous sampling
  2) Balance quick detection & cost efficiency



Hierarchical adaptive sampling in an SRS System

Research Gap

Challenges:	C1. Long-term impact of budget allocation	C2. Autonomous monitoring under dynamic budgets	C3. Hierarchical objectives & tradeoff
Typical MSPC	✗	✗	✗
Hierarchical MSPC	✗	✗	△
Typical RL	✓	✗	✗
Deep RL	✓	△	✗
Ours	✓	✓	✓

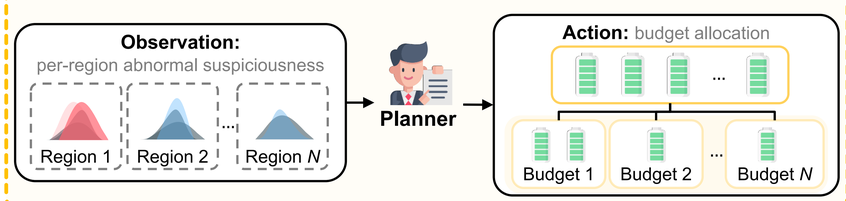
Legend: ✓ Fit △ Partial Fit ✗ Not Fit

Planning and Detecting (PAD)

To determine system status: build a statistic for each data stream

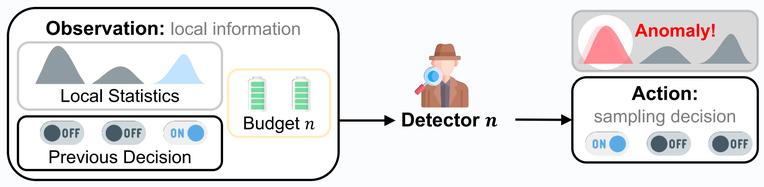
- **Planner:** Dynamic budget allocation (C1)

  Reward: negative cost of resource usage

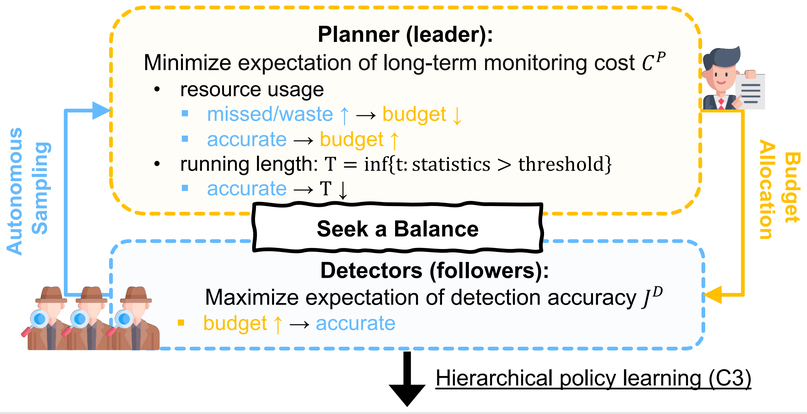


- ***N* Detectors:** Autonomous sampling under dynamic budgets (C2)

  Reward: detection accuracy (bonus if sampled, penalty if missed)



- **Leader-Follower Markov Game:** Accuracy and cost tradeoff (C3)



Hierarchical policy learning (C3)

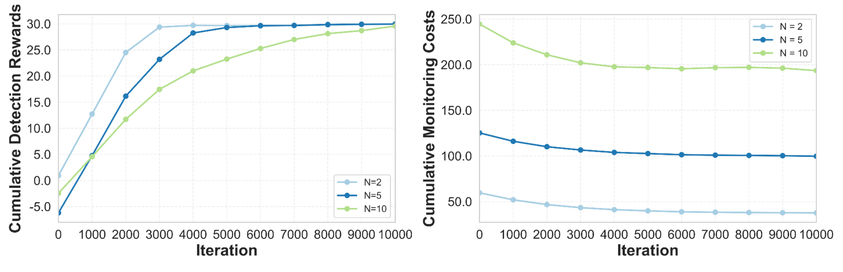
Two-Stage Policy Optimization

Stage 1: *Surrogate* planner → train *detectors* with ***MATRPO***

Stage 2: *Converged* detectors → train *planner* with ***PPO***

Theorem 1 (Detectors)
Sequential updates → J^D never decreases.

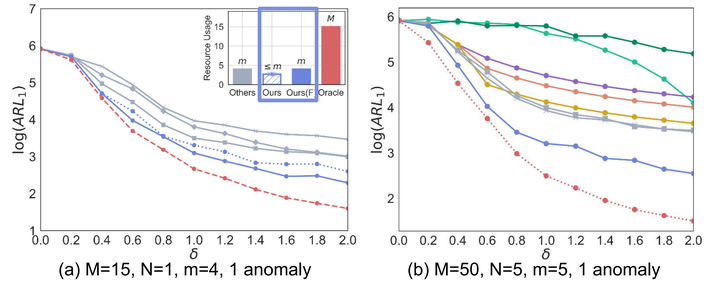
Theorem 2 (Planner)
With detectors fixed → C^P never increases.



Experiments

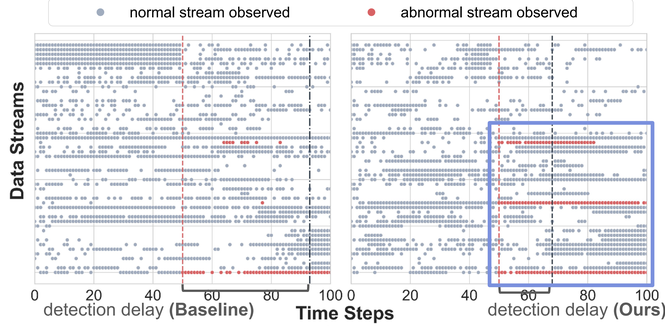
SRS Configuration		M data streams, N regions, m resources
Anomaly		mean shift of δ after $t = 50$
Metric		detection delay ARL_1 with in-control running length = 370
Baselines	(sampling)	SDQ, QUANTS, RS, Oracle
	(statistics)	CMAB, TS-BSSCD, TSSRP, NAS, AiTS
Our Variants		Ours(F), Ours

- **Detection Delay Analysis:**



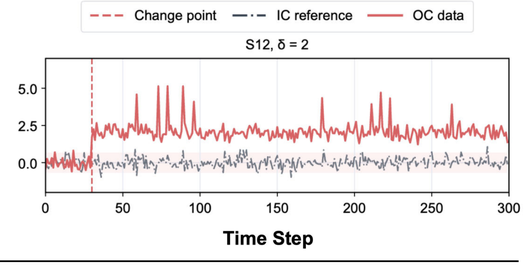
(a) M=15, N=1, m=4, 1 anomaly (b) M=50, N=5, m=5, 1 anomaly

- **Sampling Layout** (M=50, N=5, m=20, 3 anomalies):



IoT Testbed: A real-world SRS system





Structure	server → 6 Raspberry Pi 3 → 2-4 sensors per Pi 3
Data	Wi-Fi TCP, 1h warm-up, 2s interval, noise σ=0.2

